

# Sprint 2 — ROI Analysis

CS 5388 / CS 4390 Software Project Management

**Team:** 1   **Project:** Damn Parking   **Date:** October 5, 2026

## 1) Analysis Basis

This analysis examines what the limited Live Parking Monitoring slice would need to recover its costs. No customer price, adoption level, operating budget, or measured monetary benefit has been established. An actual ROI forecast cannot yet be calculated. The scenarios below are explicitly hypothetical planning tests, not research findings or agreed pricing.

Estimated one-time development labor is \$20,400–\$40,800; the expected value is \$32,280. These amounts assume 340–680 hours, expected 538, at \$60/hour. The estimated scope excludes the future 10–15-lot, 60-camera deployment. The model assumes a 12-month operating period after launch; this horizon is an analytical assumption.

## 2) Model and Cost Boundary

Let  $D$  be development cost,  $C$  be monthly operating cost, and  $R$  be monthly revenue. First-year cost =  $D + 12C$ . First-year net benefit =  $12R - (D + 12C)$ . Simple ROI =  $[12R - (D + 12C)] \div (D + 12C) \times 100\%$ . This model excludes discounting, taxes, and financing. Operating costs must include relevant support labor, hosting, data access, map fees, and maintenance without double-counting loaded development labor.

For an operator-funded subscription hypothesis,  $R = N \times P$ , where  $N$  is the number of paying operator accounts and  $P$  is the monthly fee. An account is a customer contract, not a camera or parking lot. The hypothesis needs customer validation and does not assume that any operator has agreed to pay.

## 3) Break-Even Thresholds

| Development labor  | Monthly recovery needed before operating costs |
|--------------------|------------------------------------------------|
| Low: \$20,400      | \$1,700                                        |
| Expected: \$32,280 | \$2,690                                        |
| High: \$40,800     | \$3,400                                        |

These thresholds equal  $D \div 12$ . With monthly operating cost  $C$ , required revenue becomes  $D \div 12 + C$ . Zero operating cost is only a lower-bound calculation, not an expectation of free commercial operation.

#### 4) Illustrative Subscription Scenarios

For illustration only, assume expected development cost  $D = \$32,280$  and operating cost  $C = \$500/\text{month}$ . The \$500 is a sensitivity input, not a vendor quote. First-year total cost is \$38,280 and break-even revenue is \$3,190/month. Accounts required are rounded upward.

| Hypothetical monthly fee | Accounts needed for first-year break-even |
|--------------------------|-------------------------------------------|
| \$100                    | 32                                        |
| \$200                    | 16                                        |
| \$300                    | 11                                        |

At an illustrative \$200/month per account:

| Paying accounts | Annual revenue | Net benefit | Simple ROI |
|-----------------|----------------|-------------|------------|
| 5               | \$12,000       | -\$26,280   | -68.7%     |
| 10              | \$24,000       | -\$14,280   | -37.3%     |
| 16              | \$38,400       | \$120       | 0.3%       |

The model assumes all accounts pay for all 12 months, with no churn, discounts, or collection losses. If acquisition is gradual, first-year revenue will be lower. These account counts do not establish that the limited slice can serve that scale.

#### 5) Sensitivity and Evidence Needed

At expected development cost and a \$200 monthly fee, operating costs of \$0, \$500, and \$1,000/month require 14, 16, and 19 paying accounts respectively. At \$500/month operating cost, the low-to-high development range requires 11–20 accounts. Actual costs and paid demand are therefore essential before interpreting these scenarios as attainable.

Time savings and reduced frustration may create user value, but they are not counted as cash revenue. A future social-benefit model would need measured time saved, number of affected visits, and a justified value of time, kept separate from operator cash flow.

Next evidence: intended buyer and pricing feedback; access and service quotes; recurring support effort; achievable service coverage; customer acquisition timing; and a representative accuracy/usability evaluation. ROI remains conditional until these inputs are established.